



2446 - Illuminating the AGN-Galaxy Connection with JWST: AGN Host Demographics During the Era of Galaxy Assembly

Cycle: 1, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

AGN have become key components of most galaxy evolution models, yet several open issues remain in our understanding of how the link between galaxies and their SMBHs is established and maintained. We propose to survey the host demographics of AGN during the first quarter of cosmic time in order to understand when and how the AGN-galaxy connection is established. Our study will use data from the CEERS ERS program, whose NIRCam, MIRI and NIRSpec observations will enable a wide range of AGN-related science. We will use a combination of X-ray data and CEERS MIRI and NIRSpec data to create a multiwavelength sample of AGN and use it to conduct the following science: (1) NIRCam imaging will provide rest-frame optical morphologies of AGN hosts at $z=2-6$. We will trace the emergence of bulge and disk structures and their connection to SMBH growth during the era of galaxy assembly and determine the role that galaxy mergers play in fueling AGN activity during this epoch. (2) We will use MIRI color diagnostics to perform a census of obscured SMBHs out to $z=2$, revealing the properties of this previously hidden population and determining how the fraction of obscured AGN evolves with redshift. (3) We will use NIRSpec data to conduct a survey of AGN-driven outflows at $z>3$ and measure the impact they have on their host galaxies. (4) We will search for the seeds of today's SMBHs at Cosmic Dawn by using NIRSpec to find high ionization lines from growing SMBHs at $z>7$. As part of our science analysis, we will release value-added catalogs of morphologies, emission line fluxes, and AGN identifications to supplement the CEERS dataset and directly enable future community use of JWST.

OBSERVING DESCRIPTION

There are no new observations requested as part of this archival proposal.